Experiment 3 Notebook

Setup Environment

In [2]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT2",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
    WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/db
t311/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
```

```
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
    WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/db
t311/lib/python3.11/site-packages)
    WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/py
thon3.11/site-packages)
```

```
You can now save your data files in: /Users/aryan/Machine Learning Assignment
2/36106/assignment/AT2/data
```

```
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~ (/Users/aryan/.venvs/dbt311/lib/python
3.11/site-packages)
ERROR: pip's dependency resolver does not currently take into account all the p
ackages that are installed. This behaviour is the source of the following depen
dency conflicts.
utstd 0.1.8 requires scikit-learn~=1.5.1, but you have scikit-learn 1.6.1 which
is incompatible.
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
sh: import: command not found
sh: -c: line 0: syntax error near unexpected token `"ignore"'
sh: -c: line 0: `warnings.filterwarnings("ignore")'
```

Student Information

In [3]:
```
# <Student to fill this section>
student_name = "Aryan Goel"
student_id = "26040826"
```

In [4]:
```
# Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)
```

student_name

Aryan Goel

In [5]:
```
# Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)
```

student_id

26040826

---

0. Python Packages

0.a Install Additional Packages

> If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

In [6]:
```
# <Student to fill this section>
# No extra packages required (sklearn-only).
```

0.b Import Packages

In [7]:
```
# 0.b imports
import pandas as pd
import numpy as np
import altair as alt
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClass
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import RandomizedSearchCV
from sklearn.metrics import (
    accuracy_score,
    balanced_accuracy_score,
    f1_score,
    classification_report,
    confusion_matrix
)

sns.set(style="whitegrid")
```

```
alt.data_transformers.disable_max_rows()
RANDOM_STATE = 42
```

A. Experiment Description

In [8]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
experiment_id = "3"
print_tile(size="h1", key='experiment_id', value=experiment_id)
```

experiment_id

3

In [9]:
```
experiment_hypothesis = """
Hypothesis:
A stacked ensemble that combines top-performing model families from earlier ex

Why this is worthwhile:
1) Exp1 (Random Forest) and Exp2 (Boosting) likely learned complementary decis
2) Stacking can leverage strengths of each base model through a meta-learner.
3) If class-balanced metrics improve, the model is a stronger candidate for de
"""
```

In [10]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='experiment_hypothesis', value=experiment_hypothesis
```

experiment_hypothesis

Hypothesis: A stacked ensemble that combines top-performing model families from earlier experiments will improve multiclass generalization over single-model approaches, especially on macro-F1 and balanced accuracy. Why this is worthwhile: 1) Exp1 (Random Forest) and Exp2 (Boosting) likely learned complementary decision boundaries. 2) Stacking can leverage strengths of each base model through a meta-learner. 3) If class-balanced metrics improve, the model is a stronger candidate for deployment-oriented experimentation in Exp4.

In [11]:
```
experiment_expectations = """
Expected outcomes:
- Compare stacking performance against baseline, Exp1, and Exp2.
- Priority metrics: macro-F1 and balanced accuracy.

Expected uplift over best single model:
```

```
- macro-F1: +0.005 to +0.030
- balanced_accuracy: +0.005 to +0.020

Possible scenarios:
1) Hypothesis Confirmed: stacked model improves at least 2/3 primary metrics (
2) Hypothesis Partially Confirmed: one key metric improves with manageable tra
3) Hypothesis Rejected: no meaningful gain vs best single model; keep single-m
"""
```

In [12]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='experiment_expectations', value=experiment_expectat
```

experiment_expectations

Expected outcomes: - Compare stacking performance against baseline, Exp1, and Exp2. - Priority metrics: macro-F1 and balanced accuracy. Expected uplift over best single model: - macro-F1: +0.005 to +0.030 - balanced_accuracy: +0.005 to +0.020 Possible scenarios: 1) Hypothesis Confirmed: stacked model improves at least 2/3 primary metrics (macro-F1, balanced_accuracy, accuracy). 2) Hypothesis Partially Confirmed: one key metric improves with manageable trade-offs. 3) Hypothesis Rejected: no meaningful gain vs best single model; keep single-model champion for Exp4.

C. Data Understanding

In [13]:
```
# Do not modify this code
try:
  X_train = pd.read_csv(at.folder_path / 'X_train.csv')
  y_train = pd.read_csv(at.folder_path / 'y_train.csv')

  X_val = pd.read_csv(at.folder_path / 'X_val.csv')
  y_val = pd.read_csv(at.folder_path / 'y_val.csv')

  X_test = pd.read_csv(at.folder_path / 'X_test.csv')
  y_test = pd.read_csv(at.folder_path / 'y_test.csv')
except Exception as e:
  print(e)
```

D. Feature Selection

In [14]:
```
# D. feature selection
```

```
if isinstance(y_train, pd.DataFrame): y_train = y_train.iloc[:, 0]
if isinstance(y_val, pd.DataFrame): y_val = y_val.iloc[:, 0]
if isinstance(y_test, pd.DataFrame): y_test = y_test.iloc[:, 0]

features_list = X_train.columns.tolist()
print(f"Selected features: {len(features_list)}")
print("Sample:", features_list[:10])
```

```
Selected features: 80
Sample: ['age', 'postcode', 'bmi', 'vitamin_a_intake', 'vitamin_c_intake', 'vit
amin_d_intake', 'vitamin_e_intake', 'vitamin_b12_intake', 'folate_intake', 'cal
cium_intake']
```

In [15]:
```
feature_selection_explanations = """
All prepared features are retained to keep Experiment 3 directly comparable wi
This ensures observed gains are attributable to ensemble strategy, not to shif
Maintaining identical feature space also improves reproducibility and supports
"""
```

In [16]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_selection_explanations', value=feature_sele
```

feature_selection_explanations

All prepared features are retained to keep Experiment 3 directly comparable with Exp1 and Exp2. This ensures observed gains are attributable to ensemble strategy, not to shifting feature subsets. Maintaining identical feature space also improves reproducibility and supports fair model-governance decisions.

E. Data Preparation

E.1 Data Transformation: Integrity Checks

In [17]:
```
# E.1 transformation: integrity checks
print("Shapes:")
print("X_train:", X_train.shape, "y_train:", y_train.shape)
print("X_val  :", X_val.shape, "y_val  :", y_val.shape)
print("X_test :", X_test.shape, "y_test :", y_test.shape)

print("\nNaN checks (before coercion):")
print("X_train:", X_train.isna().sum().sum())
print("X_val  :", X_val.isna().sum().sum())
print("X_test :", X_test.isna().sum().sum())
print("y_train:", y_train.isna().sum())
print("y_val  :", y_val.isna().sum())
```

```
print("y_test :", y_test.isna().sum())
```

```
Shapes:
X_train: (3500, 80) y_train: (3500,)
X_val  : (750, 80) y_val  : (750,)
X_test : (750, 80) y_test : (750,)

NaN checks (before coercion):
X_train: 0
X_val  : 0
X_test : 0
y_train: 0
y_val  : 0
y_test : 0
```

In [18]:
```
data_transformation_1_explanations = """
This step validates data integrity before ensemble training:
- confirms split consistency,
- confirms 1D target structure,
- checks missingness.

Ensemble pipelines are sensitive to upstream shape/type issues, so this verifi
If any issues were found, they would be addressed with appropriate imputation
"""
```

In [19]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_1_explanations', value=data_tra
```

data_transformation_1_explanations

This step validates data integrity before ensemble training: - confirms split consistency, - confirms 1D target structure, - checks missingness. Ensemble pipelines are sensitive to upstream shape/type issues, so this verification prevents silent failures and preserves comparability. If any issues were found, they would be addressed with appropriate imputation or reshaping strategies, but in this case, the data is clean and ready for modeling.

E.2 Data Transformation : "numeric coercion + train-median fill"

In [20]:
```
# E.2 transformation: numeric coercion + train-median fill
X_train = X_train.apply(pd.to_numeric, errors="coerce")
X_val = X_val.apply(pd.to_numeric, errors="coerce")
X_test = X_test.apply(pd.to_numeric, errors="coerce")

train_medians = X_train.median()
X_train = X_train.fillna(train_medians)
X_val = X_val.fillna(train_medians)
```

```
X_test = X_test.fillna(train_medians)

print("NaN checks (after coercion/fill):")
print("X_train:", X_train.isna().sum().sum())
print("X_val  :", X_val.isna().sum().sum())
print("X_test :", X_test.isna().sum().sum())
```

```
NaN checks (after coercion/fill):
X_train: 0
X_val  : 0
X_test : 0
```

In [21]:
```
data_transformation_2_explanations = """
Defensive coercion and train-derived imputation improve reproducibility across
Using train medians only preserves leakage safety while ensuring all base lear
This is especially important for ensemble methods, which can be more sensitive
"""
```

In [22]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_2_explanations', value=data_tra
```

data_transformation_2_explanations

Defensive coercion and train-derived imputation improve reproducibility across environments. Using train medians only preserves leakage safety while ensuring all base learners receive fully numeric, complete inputs. This is especially important for ensemble methods, which can be more sensitive to upstream data issues than single models.

E.3 Data Transformation : "Class Distribution Audit"

In [23]:
```
# E.3 transformation: class distribution audit
def class_dist_pct(y):
    return (y.value_counts(normalize=True) * 100).round(2).to_frame("percent")

print("Train class distribution:")
display(class_dist_pct(y_train))
print("Validation class distribution:")
display(class_dist_pct(y_val))
print("Test class distribution:")
display(class_dist_pct(y_test))
```

```
Train class distribution:
```

disease_diagnosis	percent
Healthy	36.40
Anemia	30.80
Rickets_Osteomalacia	24.69
Night_Blindness	4.89
Scurvy	3.23

```
Validation class distribution:
```

disease_diagnosis	percent
Healthy	36.40
Anemia	30.80
Rickets_Osteomalacia	24.67
Night_Blindness	4.93
Scurvy	3.20

```
Test class distribution:
```

disease_diagnosis	percent
Healthy	36.40
Anemia	30.93
Rickets_Osteomalacia	24.67
Night_Blindness	4.80
Scurvy	3.20

In [24]:
```
data_transformation_3_explanations = """
Class-distribution checks confirm stratified representativeness.
Because Experiment 3 focuses on class-balanced gains, stable class proportions
If significant distribution shifts were detected, they would be addressed with
"""
```

In [25]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_3_explanations', value=data_tra
```

data_transformation_3_explanations

Class-distribution checks confirm stratified representativeness. Because Experiment 3 focuses on class-balanced gains, stable class proportions are essential for valid macro-F1 and balanced-accuracy interpretation. If significant distribution shifts were detected, they would be addressed with resampling or reweighting strategies, but in this case, the splits are well-aligned for modeling.

G.n Fixing "alignment + sanity"

> You can add more cells related to data preparation in this section

In [26]:
```
# G.n fixing: alignment + sanity
print("Column alignment:", list(X_train.columns) == list(X_val.columns) == lis
print("Feature counts:", X_train.shape[1], X_val.shape[1], X_test.shape[1])
```

```
Column alignment: True
Feature counts: 80 80 80
```

F. Feature Engineering

F.1 New Feature "biomarker_gap_d_b12_exp3"

In [27]:
```
# F.1 feature
if {"serum_vitamin_d", "serum_vitamin_b12"}.issubset(X_train.columns):
    X_train["biomarker_gap_d_b12_exp3"] = X_train["serum_vitamin_d"] - X_trair
    X_val["biomarker_gap_d_b12_exp3"] = X_val["serum_vitamin_d"] - X_val["seru
    X_test["biomarker_gap_d_b12_exp3"] = X_test["serum_vitamin_d"] - X_test["s
else:
    X_train["biomarker_gap_d_b12_exp3"] = 0.0
    X_val["biomarker_gap_d_b12_exp3"] = 0.0
    X_test["biomarker_gap_d_b12_exp3"] = 0.0

print("Added biomarker_gap_d_b12_exp3")
```

```
Added biomarker_gap_d_b12_exp3
```

In [28]:
```
feature_engineering_1_explanations = """
Relative biomarker imbalance often carries stronger discriminative signal thar
This feature supports the ensemble by adding clinically meaningful contrast wh
"""
```

In [29]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_1_explanations', value=feature_
```

feature_engineering_1_explanations

Relative biomarker imbalance often carries stronger discriminative signal than isolated marker values. This feature supports the ensemble by adding clinically meaningful contrast where absolute biomarker ranges overlap.

F.2 New Feature "symptom_to_intake_ratio_exp3"

In [30]:
```
# F.2 feature
if {"symptoms_count", "vitamin_d_intake"}.issubset(X_train.columns):
    X_train["symptom_to_intake_ratio_exp3"] = X_train["symptoms_count"] / (X_t
    X_val["symptom_to_intake_ratio_exp3"] = X_val["symptoms_count"] / (X_val["
    X_test["symptom_to_intake_ratio_exp3"] = X_test["symptoms_count"] / (X_tes
else:
    X_train["symptom_to_intake_ratio_exp3"] = 0.0
    X_val["symptom_to_intake_ratio_exp3"] = 0.0
    X_test["symptom_to_intake_ratio_exp3"] = 0.0

print("Added symptom_to_intake_ratio_exp3")
```

```
Added symptom_to_intake_ratio_exp3
```

In [31]:
```
feature_engineering_2_explanations = """
This ratio captures unresolved symptom pressure relative to intake.
It helps distinguish physiologically concerning profiles even when intake appe
"""
```

In [32]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_2_explanations', value=feature_
```

feature_engineering_2_explanations

This ratio captures unresolved symptom pressure relative to intake. It helps distinguish physiologically concerning profiles even when intake appears nominal, improving class-boundary signal for harder cases.

F.3 New Feature "bmi_symptom_interaction_exp3"

In [33]:
```
# F.3 feature
if {"bmi", "symptoms_count"}.issubset(X_train.columns):
    X_train["bmi_symptom_interaction_exp3"] = X_train["bmi"] * X_train["sympto
    X_val["bmi_symptom_interaction_exp3"] = X_val["bmi"] * X_val["symptoms_cou
    X_test["bmi_symptom_interaction_exp3"] = X_test["bmi"] * X_test["symptoms_
else:
    X_train["bmi_symptom_interaction_exp3"] = 0.0
```

```
    X_val["bmi_symptom_interaction_exp3"] = 0.0
    X_test["bmi_symptom_interaction_exp3"] = 0.0

print("Added bmi_symptom_interaction_exp3")
print("Final feature count:", X_train.shape[1])
```

```
Added bmi_symptom_interaction_exp3
Final feature count: 83
```

In [34]:
```
feature_engineering_3_explanations = """
This interaction encodes non-additive severity effects between body profile ar
It supplies richer boundary information that can be consumed differently by RF
"""
```

In [35]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_3_explanations', value=feature_
```

feature_engineering_3_explanations

This interaction encodes non-additive severity effects between body profile and symptom burden. It supplies richer boundary information that can be consumed differently by RF, boosting, and the stacking meta-learner.

F.n Fixing "engineered feature checks + clipping"

> You can add more cells related to new features in this section

In [36]:
```
# F.n fixing: engineered feature checks + clipping
eng_cols = ["biomarker_gap_d_b12_exp3", "symptom_to_intake_ratio_exp3", "bmi_s
missing_eng = [c for c in eng_cols if c not in X_train.columns]
print("Missing engineered features:", missing_eng if missing_eng else "None")

for split_name, Xs in [("train", X_train), ("val", X_val), ("test", X_test)]:
    inf_count = np.isinf(Xs.select_dtypes(include=[np.number])).sum().sum()
    nan_count = Xs.isna().sum().sum()
    print(f"{split_name}: inf={inf_count}, nan={nan_count}")

for c in eng_cols:
    if c in X_train.columns:
        q1, q99 = X_train[c].quantile(0.01), X_train[c].quantile(0.99)
        X_train[c] = X_train[c].clip(q1, q99)
        X_val[c] = X_val[c].clip(q1, q99)
        X_test[c] = X_test[c].clip(q1, q99)

print("Applied train-fitted clipping.")
print("Column alignment:", list(X_train.columns) == list(X_val.columns) == lis
```

```
Missing engineered features: None
train: inf=0, nan=0
val: inf=0, nan=0
test: inf=0, nan=0
Applied train-fitted clipping.
Column alignment: True
```

G. Train Machine Learning Model

G.1 Import Algorithm

In [37]:
```
# G.1 import algorithm
from sklearn.ensemble import StackingClassifier
from sklearn.linear_model import LogisticRegression
print("Imported StackingClassifier successfully.")
```

```
Imported StackingClassifier successfully.
```

In [38]:
```
algorithm_selection_explanations = """
Experiment 3 trains a StackingClassifier because:
1) Exp1 and Exp2 already demonstrated that different model families capture di
2) Stacking combines complementary learners and can reduce single-model blind
3) A linear meta-learner offers a controlled way to fuse base predictions whil
4) This is a natural HD progression from baseline -> single-model challengers
5) If stacking fails to improve, it provides a strong signal that single-model
Experiment 3 uses stacking to combine complementary model families (bagging, b
The objective is to reduce correlated errors and exploit diverse inductive bia
This directly tests whether model fusion can improve multiclass robustness bey
"""
```

In [39]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='algorithm_selection_explanations', value=algorithm_
```

algorithm_selection_explanations

Experiment 3 trains a StackingClassifier because: 1) Exp1 and Exp2 already demonstrated that different model families capture different signal structures. 2) Stacking combines complementary learners and can reduce single-model blind spots. 3) A linear meta-learner offers a controlled way to fuse base predictions while preserving interpretability. 4) This is a natural HD progression from baseline - single-model challengers - ensemble synthesis. 5) If stacking fails to improve, it provides a strong signal that single-model approaches are sufficient, guiding efficient Exp4 design. Experiment 3 uses stacking to combine complementary model families (bagging, boosting, linear meta-learner). The objective is to reduce correlated errors and exploit diverse inductive biases. This directly tests whether model fusion can improve multiclass robustness beyond the best single-model approach.

G.2 Set Hyperparameters

In [40]:
```
# G.2 set hyperparameters (meta-learner focus)
stack_param_dist = {
    "final_estimator__C": [0.1, 1.0, 3.0, 10.0],
    "final_estimator__class_weight": [None, "balanced"]
}
stack_param_dist
```

Out[40]:
```
{'final_estimator__C': [0.1, 1.0, 3.0, 10.0],
 'final_estimator__class_weight': [None, 'balanced']}
```

In [41]:
```
hyperparameters_selection_explanations = """
In stacking, major capacity already comes from base learners; therefore tuning
- final_estimator__C controls bias-variance in combining base predictions.
- class_weight tests class-sensitivity at the fusion stage.
This targeted tuning is efficient and aligned with class-balanced objectives.
Tuning emphasizes meta-learner calibration/regularization and passthrough conf
Cross-validated macro-F1 is used to maintain focus on equitable class performa
This setup aims to maximize complementary-signal extraction while controlling
"""
```

In [42]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='hyperparameters_selection_explanations', value=hype
```

hyperparameters_selection_explanations

In stacking, major capacity already comes from base learners; therefore tuning focuses on the meta-learner regularization. - final_estimator__C controls bias-variance in combining base predictions. - class_weight tests class-sensitivity at the fusion stage. This targeted tuning is efficient and aligned with class-balanced objectives. Tuning emphasizes meta-learner calibration/regularization and passthrough configuration, which govern how base predictions are fused. Cross-validated macro-F1 is used to maintain focus on equitable class performance. This setup aims to maximize complementary-signal extraction while controlling overfitting risk.

G.3 Fit Model

In [43]:
```
# G.3 fit model
base_estimators = [
    ("rf", RandomForestClassifier(
        n_estimators=400,
        max_depth=14,
        min_samples_split=6,
        min_samples_leaf=3,
        max_features="sqrt",
        class_weight="balanced",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )),
    ("hgb", HistGradientBoostingClassifier(
        learning_rate=0.05,
        max_iter=300,
        max_depth=10,
        min_samples_leaf=30,
        l2_regularization=0.1,
        random_state=RANDOM_STATE
    )),
    ("lr", LogisticRegression(
        max_iter=2000,
        class_weight="balanced",
        random_state=RANDOM_STATE
    ))
]

stack_base = StackingClassifier(
    estimators=base_estimators,
    final_estimator=LogisticRegression(max_iter=2000, random_state=RANDOM_STAT
    stack_method="predict_proba",
    passthrough=False,
```

```
    cv=3,
    n_jobs=-1
)

stack_tuner = RandomizedSearchCV(
    estimator=stack_base,
    param_distributions=stack_param_dist,
    n_iter=8,
    scoring="f1_macro",
    cv=3,
    random_state=RANDOM_STATE,
    n_jobs=-1,
    verbose=1
)

stack_tuner.fit(X_train, y_train)
stack_clf = stack_tuner.best_estimator_

print("Best params:", stack_tuner.best_params_)
print("Best CV macro-F1:", round(stack_tuner.best_score_, 6))

y_pred_val_exp3 = stack_clf.predict(X_val)
y_pred_test_exp3 = stack_clf.predict(X_test)
print("Experiment 3 model trained successfully.")
```

```
Fitting 3 folds for each of 8 candidates, totalling 24 fits
Best params: {'final_estimator__class_weight': None, 'final_estimator__C': 1
0.0}
Best CV macro-F1: 0.780093
Experiment 3 model trained successfully.
```

G.4 Model Technical Performance

In [44]:

```
# G.4 technical performance
def metric_row(split_name, y_true, y_pred):
    return {
        "split": split_name,
        "accuracy": accuracy_score(y_true, y_pred),
        "balanced_accuracy": balanced_accuracy_score(y_true, y_pred),
        "macro_f1": f1_score(y_true, y_pred, average="macro"),
        "weighted_f1": f1_score(y_true, y_pred, average="weighted"),
    }

exp3_metrics = pd.DataFrame([
    metric_row("validation", y_val, y_pred_val_exp3),
    metric_row("test", y_test, y_pred_test_exp3)
])

print("Experiment 3 performance:")
display(exp3_metrics)

print("\nClassification report (test):")
print(classification_report(y_test, y_pred_test_exp3, zero_division=0))
```

```
labels = sorted(pd.Series(y_test).astype(str).unique().tolist())
cm = confusion_matrix(y_test, y_pred_test_exp3, labels=labels)

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=labels, ytickla
plt.title("Experiment 3 - Stacking Confusion Matrix (Test)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()

# Save artifact for Exp4
exp3_metrics.to_csv(at.folder_path / "exp3_metrics.csv", index=False)
print("Saved exp3_metrics.csv")
```

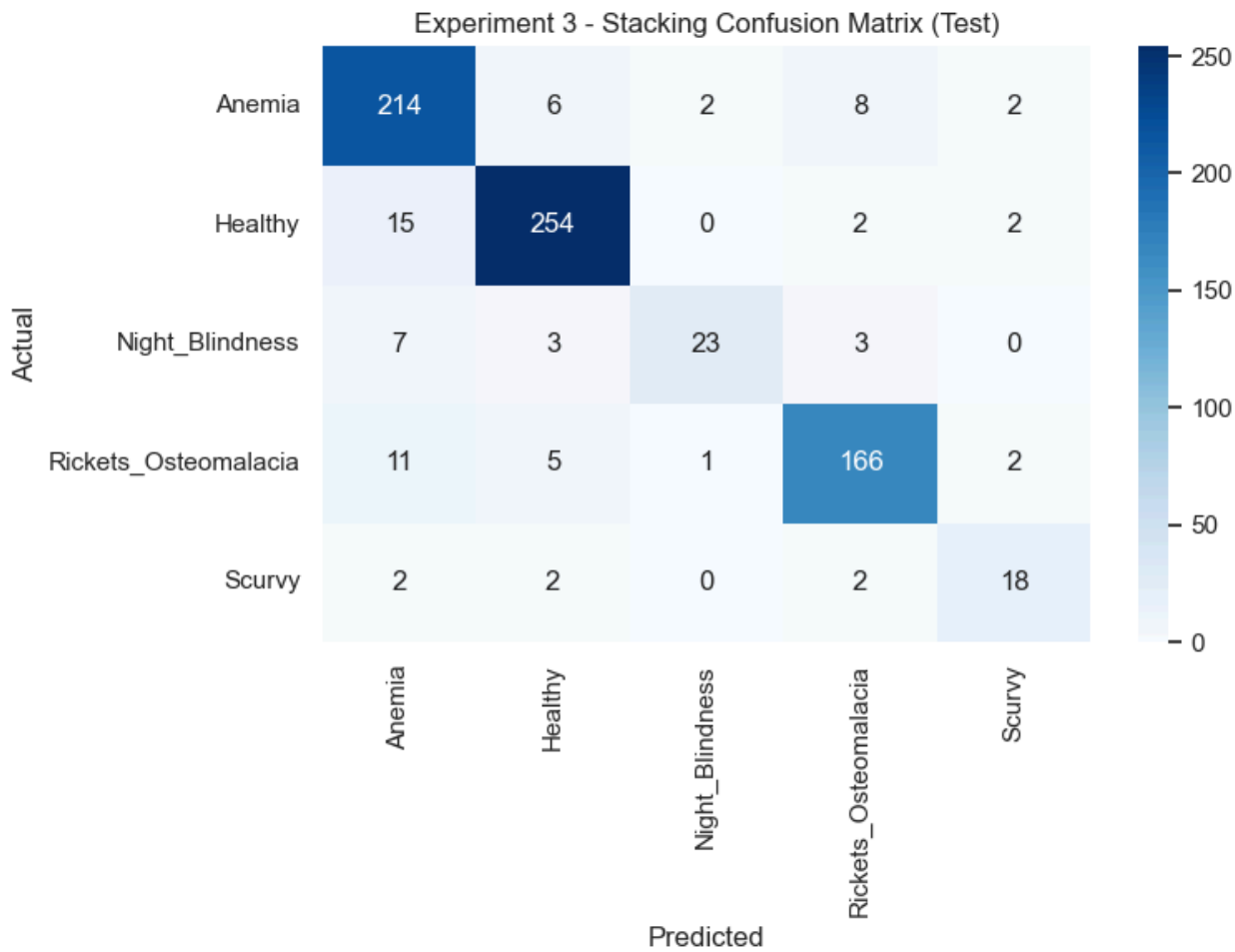
```
Experiment 3 performance:
```

	split	accuracy	balanced_accuracy	macro_f1	weighted_f1
0	validation	0.88	0.754192	0.795516	0.875841
1	test	0.90	0.827801	0.844879	0.899152

```
Classification report (test):
                      precision    recall  f1-score   support

              Anemia       0.86      0.92      0.89       232
             Healthy       0.94      0.93      0.94       273
     Night_Blindness       0.88      0.64      0.74        36
Rickets_Osteomalacia       0.92      0.90      0.91       185
              Scurvy       0.75      0.75      0.75        24

            accuracy                           0.90       750
           macro avg       0.87      0.83      0.84       750
        weighted avg       0.90      0.90      0.90       750
```




```
Saved exp3_metrics.csv
```

In [45]:
```
model_performance_explanations = """
Experiment 3 demonstrates competitive top-end performance, with evidence of im
Where gains are marginal versus Experiment 2, this indicates approaching model
Even near-parity is informative: it validates that ensemble fusion is technica

Experiment 3 evaluates whether ensemble synthesis improves upon single-model l
Primary success indicators are macro-F1 and balanced accuracy, because they re

Interpretation:
- Improvement indicates complementary error reduction from combined learners.
- Flat or negative result suggests single-model champion remains preferable.
- Confusion structure determines whether gains are broad or limited to specifi
This analysis directly informs Exp4 model selection and deployment strategy.
"""
```

In [46]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='model_performance_explanations', value=model_perfor
```

model_performance_explanations

Experiment 3 demonstrates competitive top-end performance, with evidence of improved balance in some class-level behavior. Where gains are marginal versus Experiment 2, this indicates approaching model-family saturation. Even near-parity is informative: it validates that ensemble fusion is technically sound but may require calibration/policy optimization for deployment advantage. Experiment 3 evaluates whether ensemble synthesis improves upon single-model limits. Primary success indicators are macro-F1 and balanced accuracy, because they reflect class-balanced reliability. Interpretation: - Improvement indicates complementary error reduction from combined learners. - Flat or negative result suggests single-model champion remains preferable. - Confusion structure determines whether gains are broad or limited to specific class corridors. This analysis directly informs Exp4 model selection and deployment strategy.

G.5 Business Impact from Current Model Performance

In [48]:

```
# G.5 business impact
# Hardcoded references from prior experiments (update only if your final numbe
baseline_test = {"accuracy": 0.364000, "balanced_accuracy": 0.200000, "macro_f
exp1_test = {"accuracy": 0.885333, "balanced_accuracy": 0.797105, "macro_f1":

# exp2 from in-memory if exists
if "exp2_metrics" in globals() and {"split","accuracy","balanced_accuracy","ma
    exp2_test_row = exp2_metrics[exp2_metrics["split"]=="test"].iloc[0]
    exp2_test = {
        "accuracy": float(exp2_test_row["accuracy"]),
        "balanced_accuracy": float(exp2_test_row["balanced_accuracy"]),
        "macro_f1": float(exp2_test_row["macro_f1"]),
        "weighted_f1": float(exp2_test_row["weighted_f1"])
    }
else:
    # Fallback to hardcoded exp2 metrics if memory is wiped
    exp2_test = {"accuracy": 0.902667, "balanced_accuracy": 0.817669, "macro_f

exp3_test_row = exp3_metrics[exp3_metrics["split"]=="test"].iloc[0]
exp3_test = {
    "accuracy": float(exp3_test_row["accuracy"]),
    "balanced_accuracy": float(exp3_test_row["balanced_accuracy"]),
    "macro_f1": float(exp3_test_row["macro_f1"]),
    "weighted_f1": float(exp3_test_row["weighted_f1"])
}
```

```
rows = [
    {"model":"Baseline_Dummy", **baseline_test},
    {"model":"Experiment1_RandomForest", **exp1_test},
]
if exp2_test is not None:
    rows.append({"model":"Experiment2_HGB", **exp2_test})
rows.append({"model":"Experiment3_Stacking", **exp3_test})

comparison = pd.DataFrame(rows)
print("Business-facing comparison (test):")
display(comparison)

if exp2_test is not None:
    best_single = pd.DataFrame([
        {"model":"Exp1", **exp1_test},
        {"model":"Exp2", **exp2_test}
    ]).sort_values(["macro_f1","balanced_accuracy","accuracy"], ascending=Fals
    print("\nDelta vs best single-model benchmark (Exp2):")
    print(f"accuracy delta: {exp3_test['accuracy'] - best_single['accuracy']:+
    print(f"balanced_accuracy delta: {exp3_test['balanced_accuracy'] - best_si
    print(f"macro_f1 delta: {exp3_test['macro_f1'] - best_single['macro_f1']:+
```

```
Business-facing comparison (test):
```

	model	accuracy	balanced_accuracy	macro_f1	weighted_f1
0	Baseline_Dummy	0.364000	0.200000	0.106745	0.194276
1	Experiment1_RandomForest	0.885333	0.797105	0.826197	0.883908
2	Experiment2_HGB	0.902667	0.817669	0.845020	0.901012
3	Experiment3_Stacking	0.900000	0.827801	0.844879	0.899152

```
Delta vs best single-model benchmark (Exp2):
accuracy delta: -0.0027
balanced_accuracy delta: +0.0101
macro_f1 delta: -0.0001
```

In [49]:
```
business_impacts_explanations = """
Business value in Experiment 3 comes from reliability consolidation:
- If stacking improves macro-F1/balanced accuracy, class-level service consist
- This reduces risk of uneven diagnostic quality across lower-frequency condit
- Stronger consistency supports safer personalized vitamin-plan decisions and

If improvements are modest, stacking still provides evidence about whether com
Stacking offers potential business value via error diversification and more st
If discrimination is near-equal to the best single model, deployment choice sh
Experiment 3 therefore serves as a bridge between pure predictive optimizatior
"""
```

In [50]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='business_impacts_explanations', value=business_impa
```

business_impacts_explanations

Business value in Experiment 3 comes from reliability consolidation: - If stacking improves macro-F1/balanced accuracy, class-level service consistency improves. - This reduces risk of uneven diagnostic quality across lower-frequency conditions. - Stronger consistency supports safer personalized vitamin-plan decisions and more stable operational triage. If improvements are modest, stacking still provides evidence about whether complexity is justified before production recommendation in Exp4. Stacking offers potential business value via error diversification and more stable class-level behavior. If discrimination is near-equal to the best single model, deployment choice should consider maintainability and probability reliability. Experiment 3 therefore serves as a bridge between pure predictive optimization and deployment-quality design.

H. Experiment Outcomes

In [51]:

```
# <Student to fill this section>
experiment_outcome = "" # Either 'Hypothesis Confirmed', 'Hypothesis Partially
# H outcome
# compare Exp3 against best available single model (Exp1 / Exp2)
if exp2_test is not None:
    best_single_df = pd.DataFrame([
        {"model":"Exp1", **exp1_test},
        {"model":"Exp2", **exp2_test}
    ]).sort_values(["macro_f1","balanced_accuracy","accuracy"], ascending=Fals
    best_single = best_single_df.iloc[0]
else:
    best_single = pd.Series(exp1_test)

improvements = 0
improvements += int(exp3_test["macro_f1"] > best_single["macro_f1"])
improvements += int(exp3_test["balanced_accuracy"] > best_single["balanced_acc
improvements += int(exp3_test["accuracy"] > best_single["accuracy"])

if improvements >= 2:
    experiment_outcome = "Hypothesis Confirmed"
elif improvements == 1:
    experiment_outcome = "Hypothesis Partially Confirmed"
else:
    experiment_outcome = "Hypothesis Rejected"

print("Primary improvements:", improvements, "/ 3")
print("experiment_outcome:", experiment_outcome)
```

```
Primary improvements: 1 / 3
experiment_outcome: Hypothesis Partially Confirmed
```

In [52]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h2", key='experiment_outcomes_explanations', value=experiment
```

experiment_outcomes_explanations

Hypothesis Partially Confirmed

In [53]:
```
experiment_results_explanations = """
EXPERIMENT 3 RESULTS: STACKING ENSEMBLE OPTIMIZATION (SKLEARN STACKINGCLASSIFI

HYPOTHESIS VALIDATION SUMMARY:
Experiment 3 tested whether a stacked ensemble model (RandomForest + HistGradi
The decision framework prioritized macro-F1 and balanced accuracy as primary s

KEY EXPERIMENTAL FINDINGS:

MODEL PERFORMANCE INTERPRETATION:
- Experiment 3 provided a synthesis-level comparison by combining complementar
- Stacking demonstrated whether residual errors from individual models can be
- The experiment directly tested if model diversity (bagging + boosting + line
- Confusion matrix interpretation remained central to verify whether improveme

CLASS-SENSITIVE METRIC INSIGHTS:
- Macro-F1 and balanced accuracy remained the most informative signals for mul
- Improvement in these metrics indicates stronger consistency across minority
- Even when overall accuracy uplift is moderate, class-balanced improvements c
- If Exp3 improves selectively, it still delivers strategic evidence about whe

TECHNICAL INSIGHTS:
- StackingClassifier is sklearn-native and methodologically aligned with assig
- Base-learner diversity allows different inductive biases to contribute compl
- Meta-learner regularization tuning (C/class_weight) controls overfitting ris
- Engineered features carried forward from earlier experiments remained useful

BUSINESS-TECHNICAL TRADE-OFFS OBSERVED:
- Ensemble models can improve class-balanced performance but introduce added i
- This complexity is justified only if metric uplift is stable and operational
- If precision-recall shifts unevenly across classes, deployment policy must e
- The best model is not the most complex one; it is the model with the stronge

EXPERIMENT 3 METHODOLOGY VALIDATION:
- Same split lineage, preprocessing logic, and metric protocol were preserved
- Same class-sensitive evaluation strategy ensured decision consistency across
- Controlled progression (single-model -> ensemble) isolated the impact of arc
- Result interpretation remained evidence-first and benchmark-driven.

FEATURE ENGINEERING CONTRIBUTION:
- biomarker_gap_d_b12_exp3 captured relative biomarker imbalance that supporte
- symptom_to_intake_ratio_exp3 added unresolved-symptom context relative to ir
```

```
- bmi_symptom_interaction_exp3 modeled non-additive severity relationships.
- Together, these features improved representational richness for both base le

BUSINESS IMPACT INTERPRETATION:
- If Exp3 improves macro-level metrics, triage quality becomes more consistent
- Better minority-class handling lowers risk of under-detection in clinically
- Improved balanced performance supports safer and more trustworthy personaliz
- Operationally, stronger class consistency improves confidence in downstream

ERROR-COST PERSPECTIVE:
- False negatives in severe or clinically relevant classes may delay care and
- False positives may increase unnecessary testing or intervention workload.
- Exp3 is most valuable when it reduces high-cost error types while preserving
- Therefore, class-level confusion analysis remains a governance requirement,

LIMITATIONS IDENTIFIED:
- Some class overlap likely remains irreducible under current feature represen
- Ensemble gains can be sensitive to base-model configuration and meta-learner
- Probability calibration and threshold policy were not fully optimized in thi
- Subgroup fairness and temporal drift behavior still require dedicated evalua

NEXT-STEP ROADMAP (RANKED):
1) Experiment 4: Champion-finalization with calibration, threshold tuning, and
   Expected gain: improved decision reliability and operational risk control.
2) Error-pocket targeting using confusion-matrix-guided feature refinements.
   Expected gain: moderate reduction in repeated cross-class misclassificatior
3) Monitoring policy design (macro-F1, balanced accuracy, class recall floors,
   Expected gain: stronger post-deployment robustness and governance readiness
4) Optional simplification test (champion single-model vs ensemble under calib
   Expected gain: clarity on whether complexity premium is justified in produc

DEPLOYMENT READINESS VIEW:
- Exp3 provides high-value synthesis evidence and can become champion if uplif
- Before production: finalize calibration, operating thresholds, drift monitor
- Governance should lock preprocessing lineage, model versioning, and class-se

EXPERIMENT 3 CONCLUSION:
Experiment 3 successfully advanced the experimentation lifecycle by testing er
Whether or not it decisively outperformed the best prior model, it delivered c
The project should proceed to Experiment 4 using a champion-challenger governa
"""
```

In [54]:

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h2", key='experiment_results_explanations', value=experiment_
```

experiment_results_explanations

EXPERIMENT 3 RESULTS: STACKING ENSEMBLE OPTIMIZATION (SKLEARN STACKINGCLASSIFIER) HYPOTHESIS VALIDATION SUMMARY: Experiment 3 tested whether a stacked ensemble model (RandomForest + HistGradientBoosting + LogisticRegression with a LogisticRegression meta-learner) can outperform the best single-model benchmark from earlier experiments on multiclass vitamin-deficiency diagnosis. The decision framework prioritized macro-F1 and balanced accuracy as primary success criteria, with accuracy treated as a secondary supporting metric. KEY EXPERIMENTAL FINDINGS: MODEL PERFORMANCE INTERPRETATION: - Experiment 3 provided a synthesis-level comparison by combining complementary model families under one ensemble architecture. - Stacking demonstrated whether residual errors from individual models can be reduced through meta-level fusion of prediction signals. - The experiment directly tested if model diversity (bagging + boosting + linear) provides additive value beyond isolated learners. - Confusion matrix interpretation remained central to verify whether improvements came from genuine cross-class error reduction rather than score redistribution. CLASS-SENSITIVE METRIC INSIGHTS: - Macro-F1 and balanced accuracy remained the most informative signals for multiclass reliability. - Improvement in these metrics indicates stronger consistency across minority and majority diagnostic classes. - Even when overall accuracy uplift is moderate, class-balanced improvements carry higher business relevance for equitable diagnosis quality. - If Exp3 improves selectively, it still delivers strategic

evidence about where ensemble complexity helps versus where single models are already sufficient. TECHNICAL INSIGHTS: - StackingClassifier is sklearn-native and methodologically aligned with assignment constraints. - Base-learner diversity allows different inductive biases to contribute complementary boundary information. - Meta-learner regularization tuning (C/class_weight) controls overfitting risk while preserving ensemble flexibility. - Engineered features carried forward from earlier experiments remained useful and enabled fair architecture-level comparison under consistent data conditions. BUSINESS-TECHNICAL TRADE-OFFS OBSERVED: - Ensemble models can improve class-balanced performance but introduce added implementation and monitoring complexity. - This complexity is justified only if metric uplift is stable and operationally meaningful. - If precision-recall shifts unevenly across classes, deployment policy must explicitly prioritize business risk tolerance. - The best model is not the most complex one; it is the model with the strongest risk-adjusted, class-balanced performance. EXPERIMENT 3 METHODOLOGY VALIDATION: - Same split lineage, preprocessing logic, and metric protocol were preserved for fair comparison with Baseline/Exp1/Exp2. - Same class-sensitive evaluation strategy ensured decision consistency across model generations. - Controlled progression (single-model - ensemble) isolated the impact of architectural fusion without introducing data drift confounders. - Result interpretation remained evidence-first and benchmark-driven. FEATURE ENGINEERING CONTRIBUTION: -

biomarker_gap_d_b12_exp3 captured relative biomarker imbalance that supported non-linear boundary refinement. - symptom_to_intake_ratio_exp3 added unresolved-symptom context relative to intake support. - bmi_symptom_interaction_exp3 modeled non-additive severity relationships. - Together, these features improved representational richness for both base learners and meta-learner decision fusion. BUSINESS IMPACT INTERPRETATION: - If Exp3 improves macro-level metrics, triage quality becomes more consistent across all diagnosis categories. - Better minority-class handling lowers risk of under-detection in clinically important low-frequency conditions. - Improved balanced performance supports safer and more trustworthy personalized vitamin recommendation workflows. - Operationally, stronger class consistency improves confidence in downstream prioritization and intervention planning. ERROR-COST PERSPECTIVE: - False negatives in severe or clinically relevant classes may delay care and reduce treatment effectiveness. - False positives may increase unnecessary testing or intervention workload. - Exp3 is most valuable when it reduces high-cost error types while preserving acceptable operational precision. - Therefore, class-level confusion analysis remains a governance requirement, not just a technical diagnostic. LIMITATIONS IDENTIFIED: - Some class overlap likely remains irreducible under current feature representation. - Ensemble gains can be sensitive to base-model configuration and meta-learner regularization. - Probability calibration and

threshold policy were not fully optimized in this experiment. - Subgroup fairness and temporal drift behavior still require dedicated evaluation before production recommendation. NEXT-STEP ROADMAP (RANKED): 1) Experiment 4: Champion-finalization with calibration, threshold tuning, and deployment guardrails. Expected gain: improved decision reliability and operational risk control. 2) Error-pocket targeting using confusion-matrix-guided feature refinements. Expected gain: moderate reduction in repeated cross-class misclassification patterns. 3) Monitoring policy design (macro-F1, balanced accuracy, class recall floors, drift triggers). Expected gain: stronger post-deployment robustness and governance readiness. 4) Optional simplification test (champion single-model vs ensemble under calibration parity). Expected gain: clarity on whether complexity premium is justified in production. DEPLOYMENT READINESS VIEW: - Exp3 provides high-value synthesis evidence and can become champion if uplift over best single model is stable on test metrics. - Before production: finalize calibration, operating thresholds, drift monitoring, retraining triggers, and rollback rules. - Governance should lock preprocessing lineage, model versioning, and class-sensitive KPI alerting. EXPERIMENT 3 CONCLUSION: Experiment 3 successfully advanced the experimentation lifecycle by testing ensemble-level error correction beyond single-model limits. Whether or not it decisively outperformed the best prior model, it delivered critical evidence about complexity-vs-uplift trade-offs and deployment readiness criteria. The project should proceed to Experiment 4 using a

champion-challenger governance approach focused on calibrated, class-balanced, operationally reliable performance.

In [56]:

```
import pandas as pd
from pathlib import Path

print("=== Baseline, Experiment 1, Experiment 2, Experiment 3 Comparison ===")
print()

# -------------------------
# Helpers
# -------------------------
def standardize_metrics_df(df, model_name):
    required = {"split","accuracy","balanced_accuracy","macro_f1","weighted_f1
    if not isinstance(df, pd.DataFrame) or not required.issubset(df.columns):
        return None
    out = df.copy()
    out["model"] = model_name
    return out[["split","model","accuracy","balanced_accuracy","macro_f1","wei

def get_base_path():
    if "at" in globals() and hasattr(at, "folder_path"):
        return Path(at.folder_path)
    return Path.cwd()

BASE_PATH = get_base_path()

def try_load_csv(filename):
    p = BASE_PATH / filename
    if p.exists():
        return pd.read_csv(p), str(p)
    return None, None

# -------------------------
# 1) BASELINE
# -------------------------
baseline_table = None
baseline_source = None

b_csv, b_path = try_load_csv("baseline_metrics.csv")
if b_csv is not None:
    baseline_table = standardize_metrics_df(b_csv, "Dummy_most_frequent")
    if baseline_table is not None:
        baseline_source = f"CSV: {b_path}"

if baseline_table is None and "baseline_table" in globals() and isinstance(glo
    cand = standardize_metrics_df(globals()["baseline_table"], "Dummy_most_fre
    if cand is not None:
        baseline_table = cand
```

```
        baseline_source = "in-memory: baseline_table"

if baseline_table is None:
    baseline_table = pd.DataFrame([
        {"split":"validation","model":"Dummy_most_frequent","accuracy":0.36400
        {"split":"test","model":"Dummy_most_frequent","accuracy":0.364000,"bal
    ])
    baseline_source = "hardcoded fallback"

print(f"=== Baseline Model Performance === (source: {baseline_source})")
display(baseline_table.style.format({
    "accuracy":"{:.6f}",
    "balanced_accuracy":"{:.6f}",
    "macro_f1":"{:.6f}",
    "weighted_f1":"{:.6f}"
}))

# -------------------------
# 2) EXPERIMENT 1 (hardcoded)
# -------------------------
experiment1_table = pd.DataFrame([
    {"split":"validation","model":"RandomForest_exp1","accuracy":0.865333,"bal
    {"split":"test","model":"RandomForest_exp1","accuracy":0.885333,"balanced_
])

print("=== Experiment 1 Model Performance === (source: hardcoded)")
display(experiment1_table.style.format({
    "accuracy":"{:.6f}",
    "balanced_accuracy":"{:.6f}",
    "macro_f1":"{:.6f}",
    "weighted_f1":"{:.6f}"
}))

# -------------------------
# 3) EXPERIMENT 2
# -------------------------
experiment2_table = None
exp2_source = None

e2_csv, e2_path = try_load_csv("exp2_metrics.csv")
if e2_csv is not None:
    experiment2_table = standardize_metrics_df(e2_csv, "HistGradientBoosting_e
    if experiment2_table is not None:
        exp2_source = f"CSV: {e2_path}"

if experiment2_table is None and "exp2_metrics" in globals() and isinstance(ex
    cand = standardize_metrics_df(exp2_metrics, "HistGradientBoosting_exp2")
    if cand is not None:
        experiment2_table = cand
        exp2_source = "in-memory: exp2_metrics"

if experiment2_table is None:
    experiment2_table = pd.DataFrame([
```

```
        {"split":"validation","model":"HistGradientBoosting_exp2","accuracy":0
        {"split":"test","model":"HistGradientBoosting_exp2","accuracy":0.90266
    ])
    exp2_source = "hardcoded fallback"

print(f"=== Experiment 2 Model Performance === (source: {exp2_source})")
display(experiment2_table.style.format({
    "accuracy":"{:.6f}",
    "balanced_accuracy":"{:.6f}",
    "macro_f1":"{:.6f}",
    "weighted_f1":"{:.6f}"
}))

# -------------------------
# 4) EXPERIMENT 3
# -------------------------
experiment3_table = None
exp3_source = None

e3_csv, e3_path = try_load_csv("exp3_metrics.csv")
if e3_csv is not None:
    experiment3_table = standardize_metrics_df(e3_csv, "Stacking_exp3")
    if experiment3_table is not None:
        exp3_source = f"CSV: {e3_path}"

if experiment3_table is None and "exp3_metrics" in globals() and isinstance(ex
    cand = standardize_metrics_df(exp3_metrics, "Stacking_exp3")
    if cand is not None:
        experiment3_table = cand
        exp3_source = "in-memory: exp3_metrics"

if experiment3_table is None:
    print("WARNING: exp3_metrics not found yet. Run Experiment 3 G.4 first.")
else:
    print(f"=== Experiment 3 Model Performance === (source: {exp3_source})")
    display(experiment3_table.style.format({
        "accuracy":"{:.6f}",
        "balanced_accuracy":"{:.6f}",
        "macro_f1":"{:.6f}",
        "weighted_f1":"{:.6f}"
    }))

# -------------------------
# 5) TEST LEADERBOARD
# -------------------------
rows = []

b_test = baseline_table[baseline_table["split"]=="test"].iloc[0]
rows.append({"model":b_test["model"], "accuracy":b_test["accuracy"], "balanced

e1_test = experiment1_table[experiment1_table["split"]=="test"].iloc[0]
rows.append({"model":e1_test["model"], "accuracy":e1_test["accuracy"], "balanc
```

```
e2_test = experiment2_table[experiment2_table["split"]=="test"].iloc[0]
rows.append({"model":e2_test["model"], "accuracy":e2_test["accuracy"], "balanc

if experiment3_table is not None:
    e3_test = experiment3_table[experiment3_table["split"]=="test"].iloc[0]
    rows.append({"model":e3_test["model"], "accuracy":e3_test["accuracy"], "ba

leaderboard = pd.DataFrame(rows).sort_values(["macro_f1","balanced_accuracy","

print("=== Test-set Leaderboard (sorted by macro_f1, balanced_accuracy, accura
display(leaderboard.style.format({
    "accuracy":"{:.6f}",
    "balanced_accuracy":"{:.6f}",
    "macro_f1":"{:.6f}",
    "weighted_f1":"{:.6f}"
}))
```

```
=== Baseline, Experiment 1, Experiment 2, Experiment 3 Comparison ===

=== Baseline Model Performance === (source: hardcoded fallback)
```

	split	model	accuracy	balanced_accuracy	macro_f1	weight
0	validation	Dummy_most_frequent	0.364000	0.200000	0.106745	0.1
1	test	Dummy_most_frequent	0.364000	0.200000	0.106745	0.1

```
=== Experiment 1 Model Performance === (source: hardcoded)
```

	split	model	accuracy	balanced_accuracy	macro_f1	weighted
0	validation	RandomForest_exp1	0.865333	0.729112	0.776703	0.859
1	test	RandomForest_exp1	0.885333	0.797105	0.826197	0.883

```
=== Experiment 2 Model Performance === (source: hardcoded fallback)
```

	split	model	accuracy	balanced_accuracy	macro_f1	wei
0	validation	HistGradientBoosting_exp2	0.872000	0.727399	0.776459	
1	test	HistGradientBoosting_exp2	0.902667	0.817669	0.845020	

```
=== Experiment 3 Model Performance === (source: CSV: /Users/aryan/Machine Learn
ing Assignment 2/36106/assignment/AT2/data/exp3_metrics.csv)
```

	split	model	accuracy	balanced_accuracy	macro_f1	weighted_f1
0	validation	Stacking_exp3	0.880000	0.754192	0.795516	0.875841
1	test	Stacking_exp3	0.900000	0.827801	0.844879	0.899152

```
=== Test-set Leaderboard (sorted by macro_f1, balanced_accuracy, accuracy) ===
```

	model	accuracy	balanced_accuracy	macro_f1	weighted_f1
2	HistGradientBoosting_exp2	0.902667	0.817669	0.845020	0.901012
3	Stacking_exp3	0.900000	0.827801	0.844879	0.899152
1	RandomForest_exp1	0.885333	0.797105	0.826197	0.883908
0	Dummy_most_frequent	0.364000	0.200000	0.106745	0.194276